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Exchanges for Government Bonds? Evidence During COVID-19

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Abstract. We leverage the unique institutional feature that the Israeli government bond market operates on an exchange rather than over-the-counter to analyze whether and why having an exchange affects market liquidity during a crisis. We document how the liquidity crisis in March 2020 affected the Israeli government bond market and conduct difference-in-differences analyses, comparing bid-ask spreads in exchange markets, such as the Israeli government bond market, with markets lacking an exchange. Our findings support the idea that having an exchange enhances market liquidity. A counterfactual analysis using trade data from the Israeli exchange suggests that this is due to the ability of investors to readily provide liquidity to one another and the efficient netting of trade flows on an exchange.

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Keywords: exchange • OTC market • government bonds • liquidity • crisis

1. Introduction

Treasury markets in most developed economies, especially the United States, are highly liquid markets and known to be a safe haven. This view changed, at least temporarily, in March 2020 when these markets became illiquid (Logan 2020). Central banks around the world stepped in to restore market functioning and a policy debate began on whether Treasury markets should be centralized. One proposal is to let all trades be centrally cleared and allow investors to trade with one another, as on an exchange.¹ Unfortunately, it is difficult to provide evidence for or against this reform, because Treasuries around the world trade on decentralized over-the-counter (OTC) markets. This means that we do not observe what would happen if these markets operated on an exchange.

This paper exploits the fact that there is an exception—the Israeli Treasury market, which is on an exchange—to study whether and why having an exchange enhances liquidity during a crisis. After documenting the severity of the liquidity crisis in the Israeli Treasury market, we examine the extent to which having an exchange can foster liquidity and analyze why this may be the case.

We begin our analysis in Section 3 by highlighting the unique characteristics of the Israeli government bond market compared with typical government bond markets. These markets are usually decentralized, with most trades conducted bilaterally between

counterparties or on electronic platforms that do not centrally clear trades. In some cases, a smaller portion of trades occurs on a separate exchange via a limit order book, as seen in the United States for on-the-run interdealer trades, and in South Korea, where a government bond exchange is accessible to all traders. In contrast, more than 99% of trades in Israel take place on the Tel Aviv Stock Exchange (TASE), with only a small fraction of large block trades executed off-exchange.

Building on this institutional context, we combine a novel data set of prices and trade flows, detailed in Section 4. Specifically, we gather daily bid-ask spreads (hereafter, spreads) for fixed-rate nominal government bonds in Israel from the TASE and for comparison countries, including both large and small developed economies: Germany, Greece, Ireland, Japan, Mexico, Portugal, South Korea, the United Kingdom, and the United States, from Bloomberg. Additionally, we obtain spreads from the South Korean stock exchange, the U.S. futures market from the Chicago Mercantile Exchange, and for large cap stock market indices of all countries in our sample from Bloomberg. We supplement this data with trade-flow information from TASE, which provides detailed insights into each transaction, including the type of trader (e.g., dealer-bank, mutual fund), the quantity traded, the security involved, and the transaction price.

In Section 5, we begin our empirical analysis by documenting the severity of the liquidity crisis in the

Israeli Treasury market during March 2020 as a foundation for analyzing how liquidity responded to the crisis. We highlight significant bond sell-offs driven by institutional investors, with Israeli mutual funds liquidating more than 20% of their portfolios—more than the 12% sell-off seen in the United States (as reported by Ma et al. (2022)). Despite rising yields, market depth and spreads remained relatively stable.

In Section 6, we assess how an exchange impacts liquidity, measured by daily spreads, through a series of difference-in-differences analyses. The core approach is to examine how spreads responded in markets with and without exchanges following the World Health Organization's (WHO) declaration of COVID-19 as a global pandemic. As an initial step, we compare stock market spreads (on an exchange) with government bond spreads (traded OTC) using difference-in-differences for non-Israeli countries. Then, to better control for differing liquidity crisis effects on stocks and bonds, we perform a triple-difference analysis, comparing the difference in spreads between government bonds and stocks in Israel versus other countries. We control for country-date fixed effects to address concerns of differential shocks across countries. Finally, we analyze within-country markets in the United States, Israel, and South Korea by leveraging differences between OTC and exchange trading, focusing on U.S. Treasury futures versus Treasuries and OTC versus exchange traded bonds in Israel and South Korea.

Our findings suggest that exchanges enhance liquidity during crises. The cross-country difference-in-differences analysis shows that government bond spreads in Israel increased on average by roughly 25% less than in countries where bonds trade primarily OTC. Similarly, U.S. futures market spreads rose less than U.S. government bond spreads, and spreads for exchange-traded bonds in Israel and South Korea increased less than those traded off-exchange. Although some spread differences may be driven by unobservable factors, as indicated by the variation in estimated effects across the various markets we consider, the consistent pattern in our results offers suggestive evidence in favor of exchanges.

In Section 7 we explore two key reasons why an exchange can promote liquidity. First, on an exchange everyone can trade with everyone. This is called all-to-all-trading and implies that customers can provide liquidity to one another if dealers cannot or do not want to. Second, market-wide central clearing enables instant netting of settlement obligations involving the same securities, reducing the number of securities that need to be delivered. This process lowers settlement risks and frees up balance sheet resources.

To highlight the importance of all-to-all trading, we document that significant customer-to-customer trading persisted on the Israeli exchange throughout the crisis, with these trades even surpassing dealer-to-customer

transactions at times. Notably, around 50% of the total amount investors sold was purchased by other investors, including hedge funds and algorithmic trading firms, indicating that nonmarket makers stepped in to provide liquidity. We then link customer-to-customer trading to market liquidity, hypothesizing that all-to-all trading on an exchange enhances liquidity by facilitating these transactions. We demonstrate that higher customer-to-customer trading activity is negatively correlated with bid-ask spreads, even after controlling for confounding factors such as dealer-trading activity and security-and day-specific demand and supply shocks.

To underline the benefits of efficient netting, we conduct a counterfactual analysis estimating the total settlement obligations dealers in Israel would have needed to absorb onto their balance sheets if there had been no exchange—meaning no all-to-all trading and no central clearing for all trades. This analysis complements Fleming and Keane (2021), who quantify the reduction in U.S. banks' gross settlement obligations under market-wide central netting compared with the current system without all-to-all trading, and only interdealer trades being centrally cleared. Our findings indicate that the Israeli netting rules decreased settlement obligations by 65%–81% relative to the U.S. rules—an effect of roughly similar magnitude to that found by Fleming and Keane (2021).

Moreover, because we observe a market with all-to-all trading, we can approximate what fraction of the total effect on settlement obligations comes from the impact of removing all-to-all trading, because customers who trade with other customers need to reroute their trades and trade with dealers. Our findings indicate that eliminating all-to-all trading is an important driver of the increase in settlement obligations. The substantial increase in balance sheet usage, along with recent findings by Duffie et al. (2023) showing that liquidity in the Treasury market is highly sensitive to dealers' balance sheet capacity—sharply declining once they reach around 50% of their capacity—suggests that, without an exchange, Israeli banks might have exceeded this threshold, potentially causing severe illiquidity.

Taken together, our findings have direct policy implications in shaping the ongoing policy discussion in the United States and other countries regarding whether to reform government bond markets (Benos et al. 2022). Concretely, our findings support advocates of the reform who argue that central clearing can reduce dealer balance sheet constraints and allow investors to provide liquidity to one another (Duffie 2020, Liang and Parkinson 2020).

2. Related Literature

Our main contribution is to leverage the fact that the Israeli government bond market operates on an

exchange to examine the extent to which having an exchange, rather than OTC trading, can enhance government bond market liquidity during a crisis and explain why. This adds to four strands of the literature.

First, we contribute to a growing empirical literature that studies whether to centralize OTC markets (Barclay et al. 2006, Hendershott and Madhavan 2015, Loon and Zhong 2016, Fleming et al. 2018, Biais and Green 2019, Benos et al. 2020, De Roure et al. 2020, O'Hara and Zhou 2021, Thorsten 2021, Allen and Wittwer 2023).²

Most of these papers analyze hybrid markets, in which a significant number of trades are executed bilaterally and another fraction on electronic platforms that match buyers to sellers. This market structure differs from an exchange market in that there is no all-to-all trading or central clearing for the majority of trades.

In contrast, we leverage the unique institutional feature of the Israeli bond market. This is similar to Abudy and Wohl (2018) and Plante (2017), who focus on corporate bond trading in regular times, as well as Abudy and Shust (2023), which is closer to our study. Inspired by our project, Abudy and Shust (2023) document that bid-ask spreads and trade volume rose during March 2020 in the Israeli corporate bond market. Our analysis focuses on government bonds and differs in key dimensions. We conduct a careful empirical analysis that extends beyond the Israeli market to control for unobservable country-specific factors that might drive pricing and trade behavior, including central bank interventions. Further, we bring in transaction data from the TASE to analyze why liquidity remained stable. This helps us establish a direct link between the market's design and liquidity.

Second, we complement studies that analyze different aspects of having an exchange. Most focus on eliminating counterparty risk by having a central counterparty (CCP) clearing markets for risky assets (Loon and Zhong 2014, Menkveld et al. 2015, Mancini et al. 2016, McSherry et al. 2017, Bernstein et al. 2019; for an overview of this literature, see Menkveld and Vuilleme (2021)). We focus on trading safe assets to understand the role of all-to-all trading and order-flow netting. As such, we add to the few studies that analyze netting (Bernstein et al. (2019) for stocks and Fleming and Keane (2021) for U.S. government bonds) and customer-to-customer trading (Mattmann (2021) for U.S. corporate bonds and Chaboud et al. (2022) for U.S. government bonds). Of these papers, Fleming and Keane (2021) is the closest to ours. They observe trade-level data on the U.S. government bond market and make predictions regarding what would happen if dealers could net trades with customers. We observe trades in a centrally cleared market with all-to-all

trading and make predictions about what would happen if this market cleared like it would in the United States. This allows us to highlight the importance of all-to-all trading and its effect on netting.

Third, the paper relates to a growing empirical literature that analyzes how financial regulation (such as Basel III) affects market making and, as a result, the liquidity of bond—often corporate bond—markets (Bao et al. 2018, Bessembinder et al. 2018, Dick-Nielsen and Rossi 2019, Bruche and Kuong 2021, Cohen et al. 2024).

This literature takes the market structure—typically an OTC market—as given. We focus instead on comparing market structures. This allows us to highlight the fact that regulations that affects the balance sheet can influence which market structure best promotes market liquidity.

Finally, we contribute to a young literature that analyzes events and policy responses during the COVID-19 crisis to infer lessons on how to improve the functionality of bond markets (Fleming and Ruela 2020, Pastor and Vorsatz 2020, Schrimpf et al. 2020, Eren and Wooldridge 2021, Haddad et al. 2021, Falato et al. 2021, Fleming et al. 2021, Kargar et al. 2021, Rebucci et al. 2021, Vissing-Jorgensen 2021, Benos et al. 2022, Favara et al. 2022, He et al. 2022, Ma et al. 2022).

Most of these studies focus on the United States; in contrast, we document the nature and severity of the liquidity crisis in another country (Israel). Further, we focus on the period before major policy actions were taken to avoid confounding factors and analyze how liquidity was affected by how bonds are traded: on an exchange or OTC.

3. Institutional Environment

Before conducting our empirical analysis, we describe the institutional environment and the data set we construct.

3.1. Treasury Markets and Banking Regulations

In most countries, including Israel, government bonds are issued in the primary market to a small set of banks (primary dealers). Primary dealers have a responsibility, as market makers, to buy bonds from the government and trade them with investors, brokers and one another to provide liquidity. In exchange, they enjoy benefits. For instance, in the United States, they are eligible to participate in the Federal Reserve (Fed)'s standing repo facility, overnight reverse repo facility, and securities lending program. Israel adopted this type of system in 2006 (Sade et al. 2018). Since then, designated (local and foreign) dealers are supposed to and incentivized to make markets similarly to other countries. For instance, they have exclusive access to most primary auctions and an interdealer trading platform.

The secondary market is predominantly an OTC market in all countries but Israel (see Benos et al. (2022) for an overview). Traditionally, a buyer or seller had to contact a dealer to negotiate the terms of trade bilaterally. Nowadays, dealers (who are members of a CCP) trade bilaterally or on a limit order book with one another, in which trades are cleared and netted centrally. Customers, on the other hand, have no access to the interdealer market, and most still trade bilaterally with a dealer. Only a relatively small but growing fraction of dealer-to-customer trades are executed on electronic platforms, which differ from an exchange (see Bessembinder et al. (2020) for an overview). Typically, investors run (request for quote) auctions with a selected set of dealers without having the option to trade directly with one another. Further, trades are not cleared centrally.

The lack of central clearing in the dealer-to-customer market has no direct but indirect implications for the dealers' balance sheets. A dealer may net trades of the same security with different customers for accounting purposes, for instance, when computing leverage ratios. However, the trade cannot net out for settlement purposes, as customers are not members of the CCP. This means that there is a chance that one side of the trade is delayed or fails, which introduces nonnegligible settlement risks that affect the liquidity needs of the dealer and therefore constrain the balance sheet until the settlement is complete.

Whether and how dealer balance sheets affect behavior and liquidity depends on how dealer banks are regulated. Banking regulations have been harmonized across countries over the past decades. Countries like the United States and Israel follow the global standards set by the Basel Committee on Banking Supervision.

3.2. Exchanges for Treasuries

In stark contrast to the rest of the world, Israel operates a single exchange, the TASE, for both bonds and stocks. The TASE functions as a continuous and anonymous limit order book, including opening and closing auction trading sessions, similar to large exchanges in other economies.³

For Israeli government bond trades, the TASE is the primary venue, capturing more than 99% of trades on an average day and around 77% of daily trade volume in our sample period (November 2019 to May 2020). Most of the remaining trade volume (about 17%) is executed off-exchange between dealers and customers, typically through large block trades via private negotiations, a practice aimed at minimizing price impact (Burdett and O'Hara 1987, Keim and Madhavan 1996). Although these bilateral OTC trades are electronically processed through the TASE, they are not cleared or netted by it, as the TASE does not serve as a CCP for these transactions. A small portion of

trading volume (about 6%) occurs in an interdealer market that operates on a separate limit order book, which we exclude from our analysis due to the common use of limit order books in interdealer markets globally.

Besides Israel, a handful of other countries operate sizable exchanges for government bonds (Benos et al. 2022, figure 4). The largest one is in South Korea, where the market for government bonds is hybrid with the majority of trades being traded OTC, and a smaller fraction of trade volume of roughly 40% being executed on the exchange.⁴ Moreover, in many countries, dealers trade certain types of securities, like on-the-run securities in the United States, on separate exchanges that are not accessible to investors.

3.3. Exchanges for Treasury Futures

In the United States and other countries, investors can trade Treasury futures in addition to Treasuries. Futures are standardized derivatives contracts that obligate the owner to purchase or sell a bond on a specific date and at a predetermined price. They share essentially the same fundamentals as government bonds but trade—like stocks—on exchanges; for instance, the Chicago Board of Trade.

In Israel, the futures market is small and inactive. One reason for this is that options are the preferred vehicle for investors to leverage their investments. In addition, investors use interest rate swaps to hedge their positions in government bonds.

3.4. Exchange vs. OTC Market

Exchange trading differs from traditional OTC trading in at least two key ways that can impact market liquidity.

First, on an exchange, any trader can freely and anonymously trade with others, significantly reducing the search costs common in OTC markets. In an OTC market, a customer seeking to trade with another customer must contact a dealer to find a counterparty and mediate the trade, a process known as agency trading. This method lacks anonymity, involves search frictions, and incurs a dealer markup, potentially hindering some of these trades.

Second, a CCP in exchange trading absorbs all counterparty risk and offsets trades involving the same securities, allowing them to be paired off and eliminated for accounting and settlement. Concretely, netting efficiencies arise because the CCP steps in as counterparty so trades can be netted even if a market participant has not engaged in offsetting trades with any specific counterparty. This reduces dealer settlement risks, lowers liquidity needs, and frees up balance sheet space for dealers.

4. Data

Besides Israel, we include four large developed economies, the United States, United Kingdom, Germany, and Japan, and five smaller developed countries, Portugal, Greece, Mexico, Poland, and Ireland. We choose the smaller countries because they have roughly the same amount of national debt outstanding in 2020 as Israel, as shown in Online Appendix Figure A1. Finally, we add South Korea, because it operates the largest exchange for government bonds in terms of trade volume (Benos et al. 2022).

4.1. Prices and Spreads

We obtain prices and bid-ask spreads for government bonds, futures, and stocks from February 1, 2020, until March 31, 2020 from different data sources.

We collect the daily average bid and ask Bloomberg generic prices (BGN) for the U.S., UK, German, Japanese, Polish, Portuguese, South Korean, Greek, Irish, and Mexican government bonds. The BGN bid and ask prices are calculated by Bloomberg using prices that are contributed to Bloomberg and any other information that Bloomberg considers relevant with the goal of providing “consensus” pricing. The concrete methodology is proprietary.

To cross-validate the data quality, we show in Online Appendix Figure A2 that BGN spreads for U.S. Treasuries behave similarly to spreads on TradeWeb—which is a major electronic trading platform that handles about one third of all dealer-to-customer volume in U.S. Treasuries. Existing studies have analyzed the reliability of BGN pricing data for other countries. For example, Allen and Wittwer (2023) show that the BGN midprice closely matches the prices at which Canadian dealers trade Treasuries and that institutional investors’ trade prices cluster around this midprice. Given the limited availability of bid-ask spread data in OTC markets, other researchers have also turned to Bloomberg’s pricing data to analyze bid-ask spreads in Treasury markets during 2020 (Barth and Kahn 2020, figure 12; Vissing-Jorgensen 2021, figure 2).

For Israeli government bonds, we compute daily average bid and ask prices from all prices on the limit order book of the TASE. Additionally, we collect the highest and lowest prices at which a security is traded during a day, both for exchange trades on the limit order book of the TASE and for Israeli OTC trades.

For the South Korean exchange, we calculate the daily average bid and ask prices using proprietary limit order book data provided by the exchange, similar to our approach for TASE. Given that most trade volume in South Korea occurs OTC, we primarily rely on BGN prices for the analysis, as they are intended to represent an average of the entire market. We

explicitly indicate instances where this is not the case and where we instead use the South Korean exchange data.

We restrict the sample of government bonds to those with maturities above two years that were on-the-run sometime between January 2019 and the end of our sample period to obtain conservative estimates. For Israel, this implies keeping all bonds that have a maturity above two years at any point in time between January 2019 and March 2020 because all these bonds are reissued regularly and can all be considered on-the-run.

In addition to government bond data, we obtain the daily average bid-ask spreads of all constituents of the leading large cap stock market indices for all countries in our sample from Bloomberg using the BQL interface: DAX 30 for Germany, NIKKEI 225 for Japan, FTSE 100 for the United Kingdom, S&P 500 for the United States, FTSE for Greece, ISEQ 20 for Ireland, KOSPI 200 for South Korea, MEXBOL for Mexico, WIG 20 for Poland, and PSI 20 for Portugal. For Israel, we obtain daily average bid and ask prices of the 35 largest Israeli companies—the TA 35—from the TASE.

Finally, we collect the daily (end-of day) bid and ask prices of U.S. futures contracts for the 2-year (ZT), 5-year (ZF), 10-year (TN), and 30-year (UB) prices from the Chicago Mercantile Exchange (CME).

When comparing prices or spreads in Israel to other countries, there is one complication: The business week is from Monday through Friday in the other countries and from Sunday through Thursday in Israel. To avoid dropping too many observations, we synchronize weeks by interpolating the spreads, using the values from the previous trading day if they are missing.

4.2. Trade Flows

Complementing the pricing information, we gather intraday data on the purchases and sales of Israeli government bonds by CUSIP from the TASE from November 2019 until May 2020. Each trade identifies the buyer and seller category from of the following groups: banks, dealers, foreign investors, mutual funds, pension funds, independent investment advisors, Exchange Traded Funds (ETFs), and local investors (e.g., retail investors, hedge funds, and algorithmic trading firms). We cannot identify traders individually but only know which investor group a trader belongs to.

4.3. Leverage Constraints

To analyze how capital-constrained Israeli dealer-banks were during March 2020, we collect the quarterly Basel III leverage ratio (LR)—which is known as the supplementary leverage ratio (SLR) in the United States—for all Israeli banks from the Bank of Israel from 2016q3 until 2021q4. The LR measures a bank’s

Tier 1 capital relative to its total leverage exposure.⁵ It must be larger than a specific threshold and minimally 3% according to Basel III. This puts a restriction on the balance sheet of an institution and imposes costs (CGFS 2016, Duffie 2018). As comparison, we also gather this information for all U.S. financial institutions that are required to file the “Consolidated Financial Statements for Holding Companies—FR Y-9C” from the Wharton Research Data Services (WRDS).

4.4. Data Limitation

We encounter two data limitations in our study.

First, the data on Israel’s OTC market are more restricted compared with other OTC markets, because the Israeli regulator does not collect detailed information on OTC trades. For example, the best available data lack information on trader identities and the side of the trade (buy or sell). As a result, we approximate the bid-ask spread in this market using the highest and lowest trade price of a security within a day in the spirit of Corwin and Schultz (2012) and subsequent studies.

Second we have no trade-level information on markets other than the Israeli government bond and stock market. In particular, we do not have access to Trade Reporting and Compliance Engine (TRACE) data on U.S. government bonds—which are currently available only to researchers in the Inter-Agency Working Group on Treasury Market Surveillance, such as the New York Fed or the Board—or similar trade-level data of other countries. Our solution is to rely on prior research in order to compare events in the Israeli market with those in the United States. We focus on the United States—rather than other countries that are more comparable to Israel in terms of size—because existing studies focus on the United States. In contrast, there are fewer studies for other countries, which renders a cross-country comparison more difficult.

5. COVID-19 Liquidity Crisis in Israel

Our goal is to provide evidence suggesting that liquidity in the Israeli market did not decline as sharply as in other countries because Israeli Treasuries trade on an exchange, despite the significant liquidity shock relative to the size of the economy. To build this argument, we first demonstrate the severity of the liquidity crisis in Israel and then examine how liquidity, measured through various metrics, responded to the shock.

To provide a benchmark for comparison, we draw on existing research that analyzes this episode in other countries, primarily for the United States. In doing so, we are mindful of cross-country differences, for example, the fact that the U.S. Treasury market is more

liquid than the Israeli Treasury market in regular times—see Online Appendix Tables A1 and A2 for more details.

5.1. Dash-for-Cash

Almost immediately after the WHO raised its coronavirus threat assessment to the highest level on February 28, 2020 (a Friday), financial markets plunged into a liquidity crisis in March 2020. For example, in the United States, mutual funds, hedge funds, foreign investors, and households wanted to redeem their shares of bond funds, exchange Treasuries for cash, and unwind existing positions (Barth and Kahn 2020, Cheng et al. 2020, Duffie 2020, Schrimpf et al. 2020, Eren and Wooldridge 2021, Vissing-Jorgensen 2021, Ma et al. 2022). Similar events occurred in other countries (Czech et al. 2021, Hüser et al. 2021, Moench et al. 2021).

In Israel, the triggers and severity of selling pressure in the government bond market were similar. Within a single month, fixed-income bond mutual funds had aggregate outflows of more than 20% of fund size in Israel compared with 12% in the United States (Figure 1). Aggregated over the first quarter of 2020, mutual funds, households, and foreign investors each sold roughly 2% of U.S. government debt outstanding; in Israel, pension funds, rather than foreign investors, were the largest contributor, selling more than 4% of Israeli government debt (Figure 2).

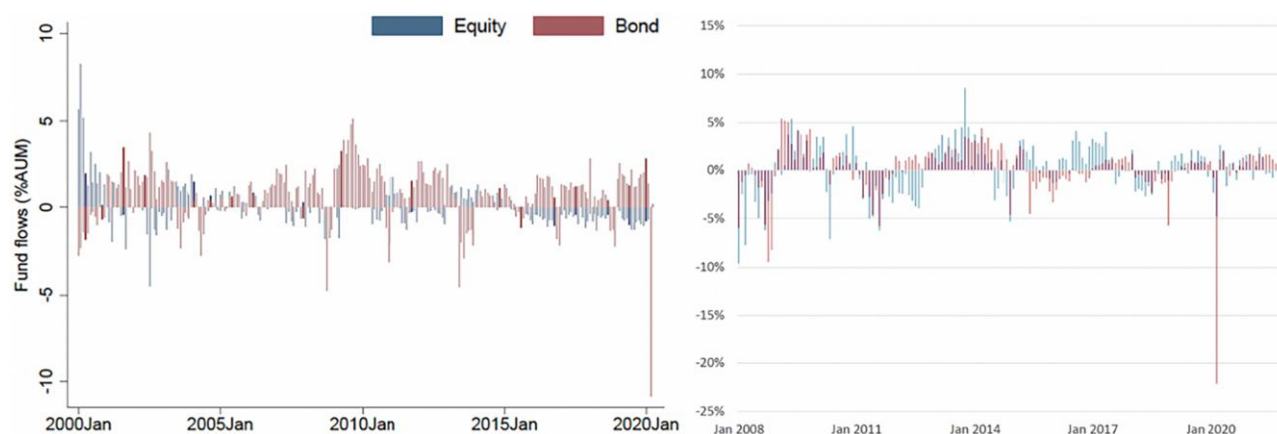
5.2. Yields, Spreads, and Market Depth

The high selling pressure impacted the prices and yields in government bond markets around the world, including Israel, as illustrated in Figure 3. This is reassuring, in that we would not expect prices or yields to remain entirely stable when demand and supply are unbalanced, even if the market mechanism is efficient.

However, in Israel, liquidity measures, including market depth, price impact, and spreads, remained relatively stable. This is shown in Figures 4 and 5 and Online Appendix Figure A3.

5.3. Leverage Constraints

One explanation for why dealer banks in OTC markets did not absorb sufficient sales on their balance sheets is that they faced overly stringent balance sheet constraints (Duffie 2020, Schrimpf et al. 2020). To show that weaker Basel III requirements or slack balance sheet constraints cannot fully explain the higher liquidity in the Israeli market (independent of market structures), we show in Figure 6 that Israeli banks faced stringent requirements and were as close, if not closer, to their LR thresholds than U.S. banks.

Figure 1. (Color online) Mutual Fund Fixed-Income Flows

Note. Monthly fund flows of bonds and equities to fixed-income mutual funds in the United States on the LHS—taken from Ma et al. (2022)—and in Israel on the RHS as the percentage of fund size (AUM).

5.4. Subsequent Policy Interventions

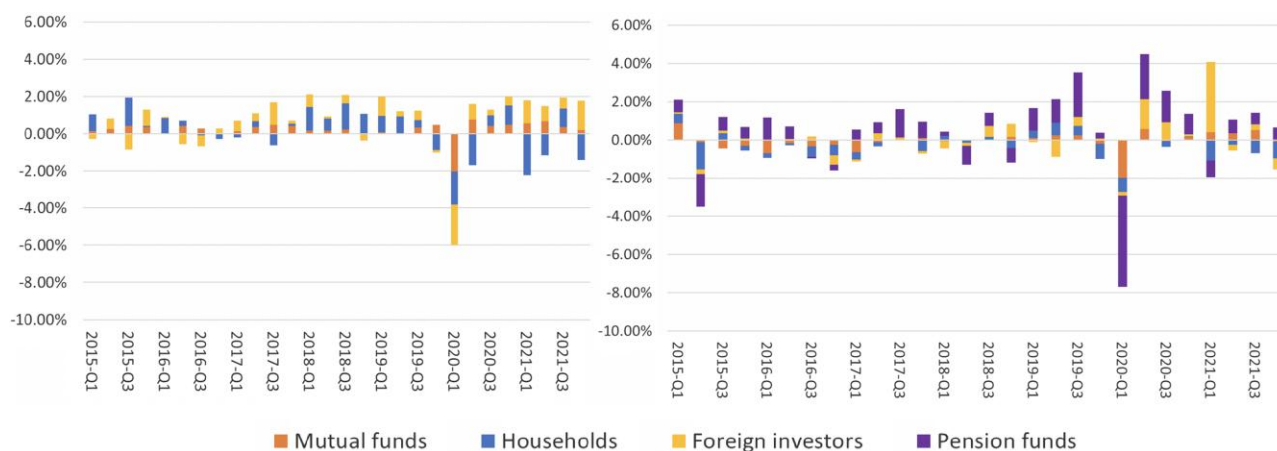
To restore market liquidity in Treasury markets, central banks around the world quickly announced policy interventions. The Bank of England, European Central Bank, Bank of Korea, Central Bank of Japan, National Bank of Poland, and Bank of Mexico began taking measures on March 11, 12, 12, 13, 16, and 20, 2020, respectively (Bank of England 2020, Bank of Japan 2020, Bank of Korea 2020, Bank of Mexico 2020, European Central Bank 2020, Hertel et al. 2022). The Fed announced that it would purchase large amounts of government bonds on March 15 and 23, 2020. In addition, to reduce balance sheet costs, government bonds were declared to be exempt from the SLR on April 1, 2020. The Bank of Israel did not exempt

government bonds from the LR but announced that it would purchase large amounts of government bonds to ensure the smooth functioning of the market on March 15 and March 23, 2020.⁶

6. To What Extent Does an Exchange Affect Liquidity?

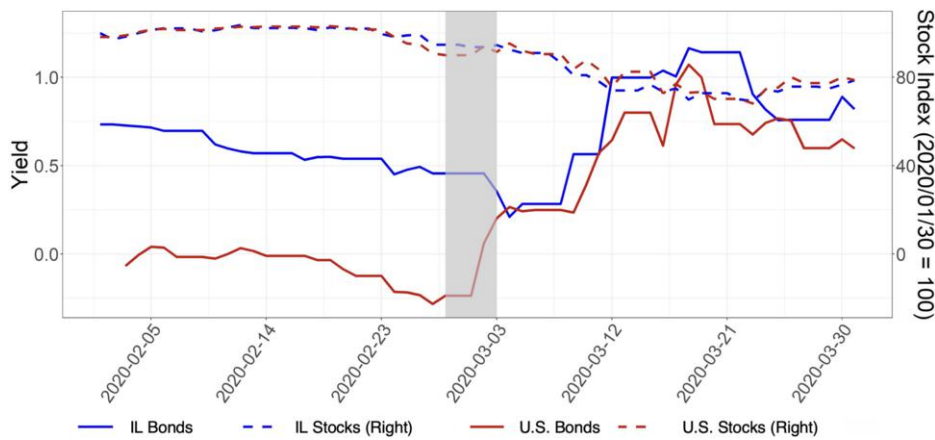
We hypothesize that exchange trading increases liquidity during a crisis.

To avoid confounding our findings by policy interventions, we focus on the time period after the WHO announced its warning on February 28, 2020, but before central banks started their interventions, which varies across countries. When we compare different

Figure 2. (Color online) Quarterly Treasury Net Transactions by Investor Type

Notes. Quarterly net purchases in the percentage of outstanding government debt of U.S. and Israeli government bonds in the secondary market by investor type. The LHS is for the United States, based on data provided by Eren and Wooldridge (2021). Investors are categorized as households, mutual funds, and foreign investors. The RHS is for Israel, for which we add pension funds.

Figure 3. (Color online) Normalized Yields of 10-Year Government Bonds and Stock Market Indices



Notes. Time series from January 1, 2020, until April 1, 2020, of the daily normalized 10-year government bond yields for Israel and the United States (left axis) and the TA 35 and S&P 500 stock market indices (right axis, January 1, 2020 = 100). We normalize the bond yields to control for cross-country differences in short-term expectations of changes in monetary policy rates. For the United States, normalized bond yields are defined as bond yield minus the yield of the next-month federal funds futures contract. For Israel, we subtract the forward rate between the three-month and one-month TELBOR rates instead, which is Israel's short-term interbank interest rate. The shaded area marks the beginning of the liquidity crisis (February 28, 2020, until March 3, 2020).

countries, we only use data until the first policy intervention, which was on March 11, 2020 (United Kingdom), to be conservative. When we use data from the same country, we use the country-specific cutoff to be more precise, but our findings are qualitatively robust when using the common cutoff date instead.

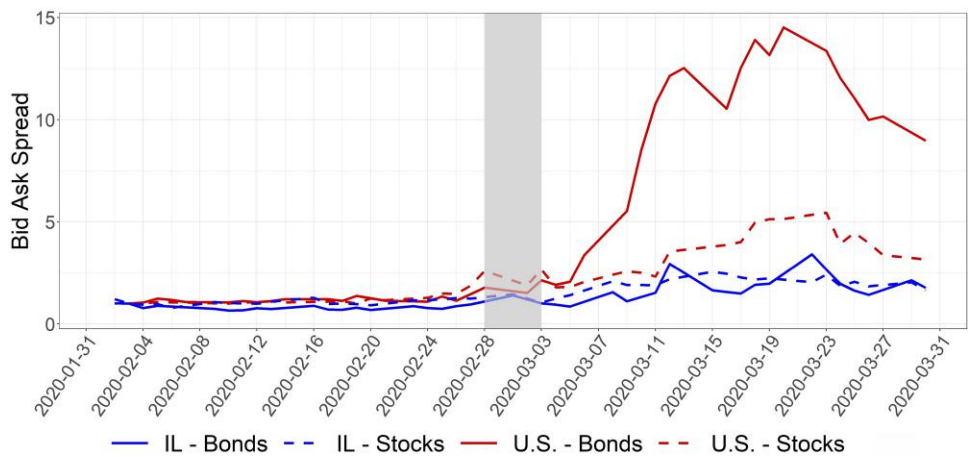
6.1. Liquidity Measure

To gather evidence for our hypothesis, we use the bid-ask spread as our main measure of liquidity for three reasons.⁷ First, this measure is ubiquitous in the literature (Amihud et al. 2012). Second, the measure can be

read from price data—unlike other liquidity measures that require trade-level information, such as trade speed or frequency; trade size; quote size; price impact coefficients (as in Online Appendix Figure A3); market depth (as in Figure 5); roundtrip transactions costs; or trade type (Fleming 2003, Feldhütter 2012, Kargar et al. 2021). Third, vast theoretic work explains why the spread soars when liquidity evaporates (Glosten and Milgrom 1985, Bessembinder and Venkataraman 2010).

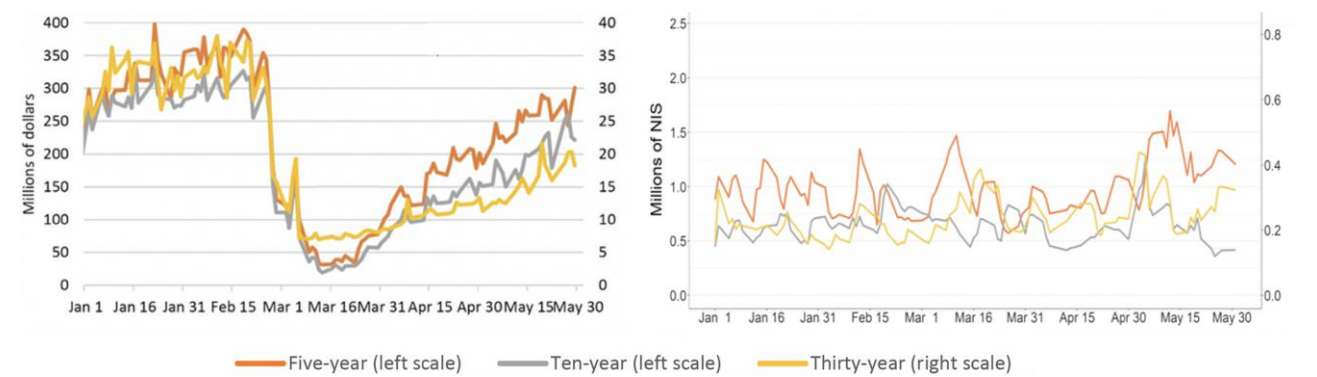
We express all spreads as a percentage of the midpoint price. Alternatively, we could use the difference

Figure 4. (Color online) Time Series of Spreads of Government Bonds and Stock Indices



Notes. Average daily bid-ask spreads of Israeli and U.S. government bonds, the TA 35 index, and the S&P 500 from February 1, 2020, until March 31, 2020. Spreads on February 1, 2020, are normalized to one. When averaging bond spreads, we weight by the bonds' notional amounts. The shaded area marks the beginning of the liquidity crisis (February 28, 2020, until March 3, 2020).

Figure 5. (Color online) Market Depth



Notes. Depth of the limit order book in the interdealer market for on-the-run U.S. government bonds on the LHS and for the Israeli government bond market on the RHS. For the United States, the graph is taken from Fleming et al. (2021). We replicate it for Israel to the best of our ability. We first average the order amount associated with the best bid and the amount associated with the best ask for a security each time the order book changes. We then average over the course of the day to obtain the daily average market depth of a security. Finally, to obtain a daily market depth measure per maturity category, we average over securities in the same maturity category (e.g., 5, 10, 30 years), weighted by the amount outstanding of each security. Market depth is measured in million U.S. dollars and NIS, respectively.

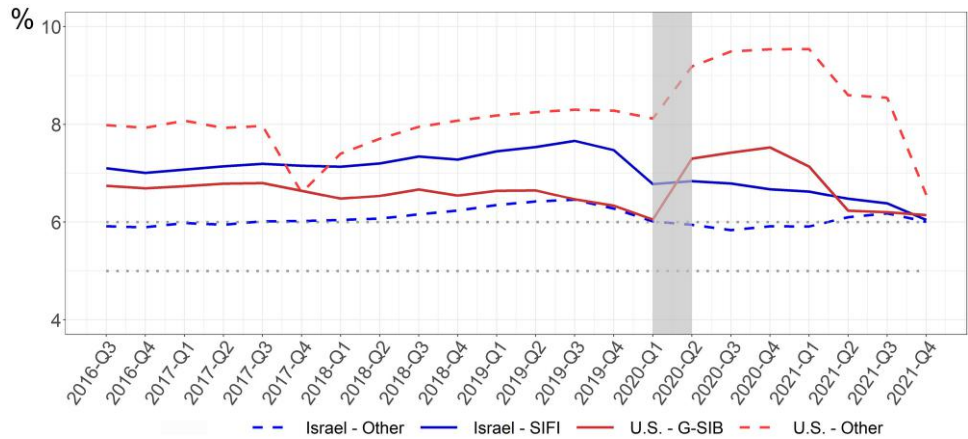
between the bid and ask price without normalizing by the midpoint price; the findings remain robust. Spread summary statistics are provided in Online Appendix Tables A1 and A2. In all regressions, spreads are in logs. We therefore need to convert all coefficients and interpret them as percentages: $100 \times [\exp(\text{coefficient}) - 1]$ is the percentage change in the spread.⁸ We present the estimation results of our preferred fixed effects specifications in the main text and report the corresponding ordinary least squares (OLS) specifications in the Online Appendix.

6.2. Warm-up

A first approach to test our hypothesis is to compare spreads in the stock market with spreads in the government

bond market in a difference-in-differences (DD) regression. For countries other than Israel, this could tell us by how much liquidity in the stock market differed from liquidity in the government bond market, because stocks are traded on an exchange and bonds are not, if we are willing to assume that the liquidity crisis would have had the same effect on stock and bond spreads if both were traded on an exchange. This is a strong assumption, because bonds and stocks differ in many dimensions and the nature of the dash-for-cash.⁹ We therefore do not interpret the DD estimation results as causal; instead, we discuss them as groundwork for our main regression: a triple-difference (DDD) analysis.

Figure 6. (Color online) Time Series of the LR of an Average Bank in the United States and Israel



Notes. Time series of the quarterly reported LR (in %) of an average bank on the consolidated basis of two types that face different leverage ratio thresholds in the United States and Israel from 2017q1 until 2021q3. In Israel there are two systemically important banks (SIFI) banks and six non-SIFI banks that face the 6% and 5% leverage constraint, respectively. In the United States, there are eight global systemically important banks (G-SIBs) that face the 5% threshold on the consolidated basis and 14 other institutions that report the consolidated financial statements for holding companies—FR Y-9C—and face the 3% threshold. We exclude an outlier, DWS USA corporation, given the huge LR levels.

We estimate two specifications with data from all countries in our sample but Israel. In the first, we regress the log of the daily spread of security i —for instance, the U.S. Apple stock or the U.S. two-year government bond— $\log \text{BAS}_{it}$, on a country fixed effects, ξ_{ct} , and an indicator variable, post , that equals one in the crisis period and an indicator variable, stock , for whether the security is a stock. In the second preferred specification, we add a security fixed effect and country-date fixed effects, ξ_i and ξ_{ct} , respectively:

$$\log \text{BAS}_{it} = \xi_c + \beta_1 \text{stock} + \beta_2 \text{post} + \beta_3 \text{stock} \times \text{post} + u_{it}, \quad (1)$$

$$\log \text{BAS}_{it} = \xi_{ct} + \xi_i + \beta_3 \text{stock} \times \text{post} + u_{it}. \quad (2)$$

In line with our expectation, Table 1 shows that the coefficient β_3 is negative. In contrast, when we run these regressions using proprietary data from the Israeli and South Korean exchanges as a placebo test, the coefficients are statistically insignificant.

Online Appendix Table A4 presents the β_3 estimates when estimating the fixed effects specification separately for each country rather than pooling the data and including a country-day fixed effect. Nearly all country-specific estimates are negative, with the exception of Poland.

In Poland, stock spreads increased more than bond spreads, even though stocks are exchange-traded, whereas Polish government bonds are primarily traded OTC. Although it is difficult to be certain, given the limited research available on Poland, one possible explanation is that the government bond market there did not experience a significant dash-for-cash. This could explain why Polish government bond yields remained more stable and trading volumes did not spike as sharply as in other countries (as seen in charts 1 and 2 of Polish Ministry of Finance (2020)). Another possibility is that the Polish OTC market

functioned more smoothly compared with their stock market than other OTC markets, leading to better liquidity despite comparable selling pressures. Although we lack trade-flow data to confirm this and existing research is sparse, it seems unlikely, given that Polish OTC market rules do not appear to differ significantly from those of other OTC markets.

6.3. Cross-Country Analyses

With data from Israel, where bonds are traded on an exchange, we can better isolate what effect having an exchange has on spreads because we can control for any fundamental difference between stocks and government bonds. Exploiting variation in the spreads of stocks and government bonds in Israel vs. the other countries, we run the following DDD specification:

$$\log \text{BAS}_{it} = \xi_{ct} + \xi_i + \delta_1 \text{bond} \times \text{post} + \delta_2 \text{bond} \times \text{IL} \times \text{post} + u_{it}. \quad (3)$$

In our preferred specification, shown in Table 2, we include security fixed effects, ξ_i , and country-date fixed effect, ξ_{ct} , to absorb unobservable factors, including different severity and time-varying of the dash-for-cash shock, between the countries. The main coefficient of interest, δ_2 , is identified based on how daily spreads of bonds compared with stocks change in Israel versus the other countries after February 28, 2020, relative to how they were before, once we exclude time-country-specific trends that are common to all securities. Online Appendix Table A5 shows the analogous regression with only country fixed effects.

We first estimate both regressions using data from all countries: We find that the government bonds' spread over stocks rose by about $100 \times [\exp(-0.317) - 1] \approx 27\%$ less in Israel than in the other countries.¹⁰ Next, we divide the sample into large and small economies. For large economies, we re-estimate the regressions with and without the United States, as its bond market might have been more affected by the shock relative to its stock market, given the status of U.S. Treasuries as the safest in the world. For small economies, we re-estimate the regressions with and without Poland, given that our earlier analysis identified it as an outlier in the sample. The estimation coefficient is similar for these two cases and imply an effect of roughly 24%.

We could interpret these estimation findings as causal, meaning that the Israeli government bond spreads were lower because of the exchange, if spread pretrends were parallel and any effect the dash-for-cash had on the bond-stock difference (unrelated to the trading mechanism) was the same in both countries. Importantly, the magnitude of the dash-for-cash can differ across countries thanks to the country-date fixed effect.

Table 1. Within-Country DD of Bonds vs. Stocks

	OTC	TASE	KRX
$\text{stock} \times \text{post}$	−0.171*** (0.016)	+0.061 (0.057)	+0.058 (0.059)
Observations	48,284	1,831	7,169
Adjusted R^2	0.943	0.925	0.814

Notes. Results of DD Regression (2) with country-day and security fixed effects. Online Appendix Table A3 presents the estimation results of the analogous regressions with only country fixed effects. In column OTC, we use data from all countries in our sample excluding Israel from February 1, 2020, through March 10, 2020, inclusive. As placebo tests, we show the estimates for the Israeli and South Korean Stock Exchange in columns TASE and KRX, respectively. Here, we where we use proprietary price data from the two exchanges, again from February 1, 2020, until each country's first announcement of monetary policy measures. The dependent variable is the daily log spread of a security; stock is one for stocks and zero for bonds; post is one starting on February 28, 2020. Standard errors are clustered by security.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table 2. Cross-Country DDD

	All	Large	No United States	Small	No Poland
$bond \times post$	+0.171*** (0.016)	+0.217*** (0.017)	+0.120*** (0.021)	+0.055 (0.036)	+0.128*** (0.039)
$bond \times IL \times post$	−0.319*** (0.063)	−0.365*** (0.063)	−0.268*** (0.065)	−0.204*** (0.071)	−0.277*** (0.073)
Observations	49,962	36,503	17,624	15,137	13,949
Adjusted R^2	0.942	0.926	0.939	0.897	0.897

Notes. Results of DDD Regression (3), which includes security and country-day fixed effects. Online Appendix Table A5 shows the results of the corresponding regressions with only country fixed effects. In column All, we use data of all countries in our sample from February 1, 2020, through March 10, 2020, inclusive. In column Large, we compare the large economies, United States, United Kingdom, Germany, and Japan, to Israel, and in column Small the small economies, Greece, Ireland, Mexico, Poland, Portugal, and South Korea. We exclude the United States from the large countries in column No United States and Poland from the small countries in column No Poland. The dependent variable is the daily log spread of a security. Indicator variable bond is one for bonds and zero for stocks; IL is one for Israeli securities and zero otherwise; post is one starting on February 28, 2020. Standard errors are clustered by security.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To test for pretrends, we perform a dynamic DDD analysis for our preferred regression specification with country-date fixed effects (Figure 7). We find that, in Israel, government bond spreads did not diverge from stock spreads, compared with other countries, before the dash-for-cash began but declined afterward. At its peak, the estimated coefficients imply a large effect of above 50%. The effect was not immediately on Friday, February 28, when the WHO raised the coronavirus threat to its highest level. The delay might be due to dealers initially managing selling pressure effectively. However, as their inventories filled up, liquidity declined. Data show this shift

occurred in early March (Duffie 2020), aligning with spread changes. Spreads rebounded when central banks intervened on March 11. Volatility remained high as more interventions and pandemic news unfolded until late April, when spreads stabilized slightly below precrisis levels.

6.4. Within-Country Analyses

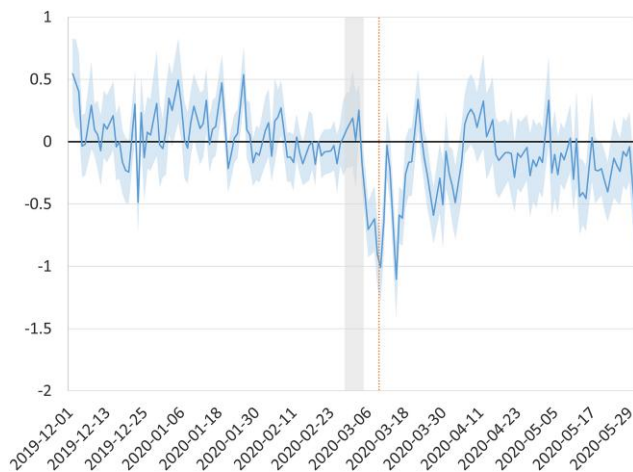
Although our main DDD analysis includes country-day fixed effects, unobservable differences across countries could still influence some of the previous findings. The primary concern is that Israel's government bond market experienced a milder shock relative to its stock market compared with other countries. To complement the across-country analysis, we end our analysis with comparisons of markets within the same country, focusing on the United States, Israel, and South Korea.

In all cases, we estimate DD regressions of the following form:

$$\log BAS_{it} = \beta_0 + \beta_1 \text{exchange} + \beta_2 \text{post} + \beta_3 \text{exchange} \times \text{post} + u_{it}, \quad (4)$$

where BAS_{it} is the bid-ask spread of security i on day t , indicator variables exchange is one if the security i is traded on the exchange, and post is one starting on February 28, 2020. In our preferred specifications, we include security and day fixed effects ξ_i and ξ_{it} .¹¹

For the United States, we compare U.S. Treasury futures with U.S. government bonds, leveraging the fact that both assets have essentially the same fundamentals but one is traded on an exchange and the other is not. The estimates of the static DD are reported in Table 3; Figure 8 shows the dynamic analysis. The estimates suggest that the futures spread rose, on average, by 66% less than the government bond spread, which is larger than our previous estimates. We conjecture that some of the estimated effect

Figure 7. (Color online) Dynamic Cross-Country DDD

Notes. The $\delta_{2,k}$ coefficients and the 95% confidence bounds of regression: $\log BAS_{it} = \xi_i + \xi_{it} + \sum_{k=1}^{182} \delta_{1,k} \text{bond} \times \text{day}_k + \sum_{k=1}^{182} \delta_{2,k} \text{bond} \times \text{IL} \times \text{day}_k + u_{it}$, where BAS_{it} is the spread of security i on day t ; day_k equals one on the k th day from December 2, 2019, until May 31, 2020, and zero otherwise; bond is one for government bonds and zero for stocks; ξ_i is a security; and ξ_{it} is a country-date fixed effect. The baseline is February 19, 2020. The gray area shades the period between February 28, 2020, and March 3, 2020. The vertical dashed line marks the last day before the first policy intervention, March 10, 2020.

Table 3. Within-Country DD: United States, Israel, and South Korea

	United States	Israel	South Korea
$exchange \times post$	−1.073*** (0.084)	−0.761** (0.353)	−0.318** (0.112)
$exchange$		+1.130*** (0.189)	+0.233 (0.212)
Observations	2,643	496	878
Adjusted R^2	0.932	0.788	0.659

Notes. Results of DD Regression (4) but adding security, and day fixed effects for the United States, for Israel, and for South Korea, using data from February 1, 2020, through the first day of each country's central bank intervention. Online Appendix Table A7 shows the analogue estimates for the regression without fixed effects. For the United States, the dependent variable is the daily log spread of a security; exchange is one for futures and zero for bonds. For Israel and the South Korean exchange, exchange is one when the trade is on the exchange and zero when it is OTC. For the United States, where the same security does not trade both on and off the exchange, the exchange variable is omitted given perfect collinearity with the fixed effects. For Israel, the dependent variable is the log of the difference between the highest and lowest trade price of a security within a day, normalized by the average of these two prices. We exclude cases when there is a single trade of a security within a day, or the highest and lowest trade prices are identical (likely, because all investors execute trades in the same direction). For South Korea, we use the daily bid-ask spread from the South Korean stock exchange for trades on the exchange and the Bloomberg bid-ask spread for all trades on and off the exchange. In all specifications, post is 1 starting on February 28, 2020. Standard errors are clustered by security.

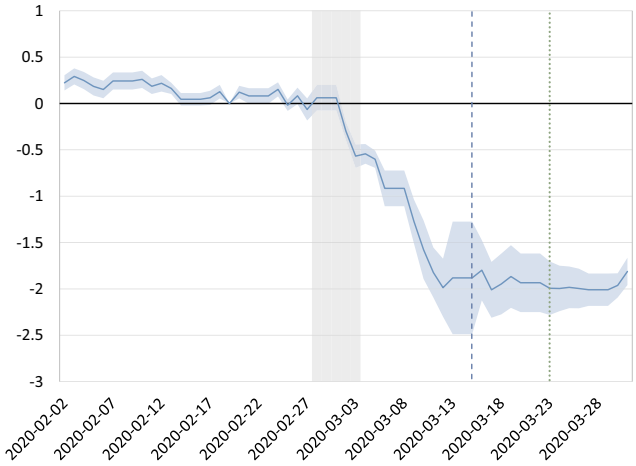
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

comes from the higher selling pressure in the Treasury cash market relative to the futures market in light of research that suggests that the Treasury market might have faced larger selling pressure than the futures market (Barth and Kahn 2020, Schrimpf et al. 2020).¹²

Lastly, for South Korea and Israel, where a portion of government bond trades occur on an exchange, we can conduct DD analyses comparing bonds traded on versus off the exchange within each country. The advantage of this approach is that it allows us to focus on the same asset class within each country, addressing concerns that arise when comparing across countries and asset classes. The main conceptual disadvantage is that we cannot control for endogenous selection between trading methods that can drive unobserved differences.

Moreover, there are two practical disadvantages. First, we need to rely on proxy variables to measure the bid-ask spread in the OTC market, given that we do not observe OTC trades. For Israel, we use the high-low spread (the difference between the highest and lowest trade price of a security within a day, normalized by the average of these two prices). For South Korea, we use the Bloomberg spread that relies on averages, which are likely driven by OTC trades given that the majority of trades are traded off the exchange

Figure 8. (Color online) Dynamic DD Analysis: U.S. Treasury vs. U.S. Futures



Notes. The $\beta_{3,k}$ coefficients and the 95% confidence bounds of regression: $\log BAS_{it} = \xi_i + \xi_t + \sum_{k=1}^{124} \beta_{3,k} \text{future} \times \text{day}_k + u_{it}$, where BAS_{it} is the spread of security i on day t ; day_k equals one on the k th day from January 9, 2020, until May 31, 2020, and zero otherwise; future is one for U.S. Treasury futures and zero for U.S. government bonds; ξ_i is a security; and ξ_t is a date fixed effect. The baseline is February 19, 2020. The gray area shades the period between February 28, 2020, and March 3, 2020. The vertical dashed lines correspond to the Fed's interventions on March 15 and 23, 2020, respectively.

in South Korea. Second, the sample size is relatively small for both countries. For Israel, this is due to the limited number of bonds that trade OTC given the small share of the market. For South Korea, the issue arises because some bonds predominantly trade either exclusively on the exchange or OTC. For example, less liquid bonds typically trade off the exchange and are excluded when we apply security fixed effects. These and similar observations are absorbed by the security fixed effect rather than helping to identify our coefficient of interest. Therefore, there is not enough statistical power to conduct the dynamic analyses.

Despite lower statistical power and data limitations, we estimate coefficients that are in line with our hypothesis for both countries. In Israel, spreads on the exchange rose on average by 53% and in South Korea by 27% less than spreads in the OTC market (Table 3).¹³

6.5. Take Away

We presented a series of complementary regression specifications to support the hypothesis that exchange trading increases liquidity during a crisis compared with over-the-counter markets. The extent to which an exchange reduces spreads during the crisis varies, depending on the specific market analyzed, being roughly 25% on average and more than 50% at the peak.

None of the analyses alone can establish causality definitively. In the cross-country analysis, we are concerned that unobservable differences across countries, not captured by the country-day fixed effects, could affect the results. In the within-United States analysis, which compares the cash and futures markets, the worry shifts to differences across these markets, such as varying trader types or purposes. Finally, in the within-South Korea and within-Israel analysis, which compares the same asset within each country, the concern is unobservable differences across market segments, like trade size or trader type, influencing market choice.

However, taken together, our analyses produce suggestive evidence that points in the same direction: Trading government bonds on an exchange can increase liquidity during a crisis period. Next, we discuss and analyze why.

7. Why Does an Exchange Promote Liquidity?

An exchange may promote liquidity for two main reasons. First, everyone can anonymously trade with everyone. This implies that customers can provide liquidity for one another if dealers cannot or do not want to. Second, a CCP absorbs any counterparty and settlement risk by netting trades. Naturally, counterparty risk is lower for government bonds than for stocks because trades settle quickly and involve a single transaction of a safe asset (Duffie 2020). However, some counterparty risk can build up when trades fail to settle as planned, especially during times of distress. Then the settlement date is moved forward until the security is ultimately delivered (Fleming and Keane 2021).

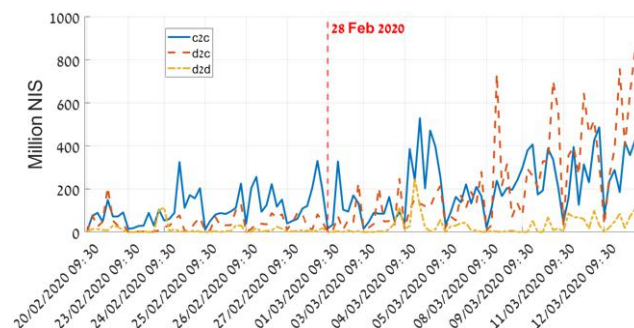
7.1. All-to-All Trading

We begin by documenting that customers traded significant volumes with each other on the Israeli exchange, with the key takeaway being that customer-to-customer trading remained substantial throughout the crisis.

We see this in Figure 9, which visualizes the hourly trade flow between dealers, between dealers and customers, and between customers. During the crisis period, between February 28, 2020, and March 13, 2020, 56% (51%) of the amount customers bought in an average hour (day) was sold to other customers, that is, the average ratio between the amount bought by customers through customer-to-customer transactions in an hour over the total amount customers bought in that hour is 56%.

An alternative scenario could have been a complete drying up of customer-to-customer trading during the crisis, leaving dealers as the only buyers of Treasuries, but our findings suggest otherwise.

Figure 9. (Color online) Hourly Trade Volume per Market Segment



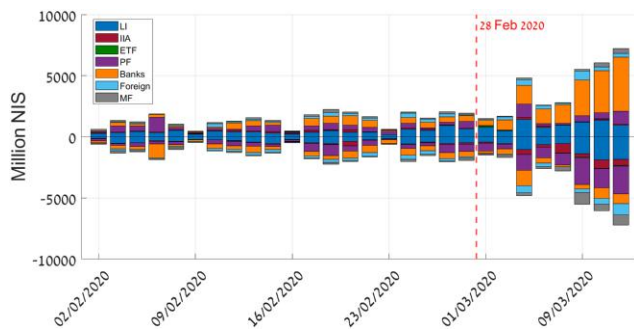
Notes. Hourly buying volume of customers from customers (c2c, the solid line), dealers from customers (d2c, the dotted line), and dealers from dealers (d2d, the dashed line) in million NIS from February 20, 2020 until March 13, 2020. The vertical line marks the beginning of the crisis period, February 28, 2020, a Friday, where no trades took place in Israel. Fleming and Keane (2021) show the analogous graph for the U.S. market at the daily-level.

To identify which types of traders leaned against the wind and bought government debt during the crisis, we analyze daily trade-flow data from TASE, where we can separate the following types: banks, dealers, foreign investors (FIs), mutual funds (MFs), pension funds (PFs), independent investment advisors (IIAs), ETFs, and local investors (LIs), which encompass retail traders, hedge funds, and algorithmic trading firms. Figure 10 shows the daily purchases and sales by each trader group from February 2, 2020, to March 13, 2020. If only dealer banks had been buying while nondealers were selling during the crisis, all upward bars would appear in orange—the dealer's color. Although we see that dealers played an important role in taking the buy side, the multicolored positive bars indicate that nondealers also made purchases.

In total, roughly 43% of the total amount sold by customers during the crisis period was bought by nondealers. Local investors—which include hedge funds and algorithmic and high-frequency trading firms—bought the largest share among nondealers, with roughly 24%, whereas pension funds, mutual funds, and foreign investors purchased less. Moreover, local investors were net buyers during a substantial part of the crisis period (until March 8, 2020), meaning that, on the aggregate, local investors bought more than they sold in a day. This suggests that allowing sophisticated traders, such as algorithmic trading firms, to trade directly with investors may help balance unbalanced demand and sell sides during times of crisis.

We link customer-to-customer trading to market liquidity, with our hypothesis being that all-to-all trading on an exchange enhances liquidity by facilitating these customer-to-customer transactions, to complement

Figure 10. (Color online) Daily Trade Flows by Investor Group



Notes. Total amount bought (the positive bars) and the total amount sold (the negative bars) of Israeli government bonds in a day by trader type—that is, banks (who are dealers), local investors (LI), mutual funds (MF), pension funds (PF), independent investment advisors (IIA), ETFs, and foreign investors (Foreign)—from February 20, 2020, until March 13, 2020. The vertical line marks the beginning of the crisis period, February 28, 2020, a Friday, where no trades took place in Israel. Trade flows are measured in million NIS.

the evidence we provide in Section 6. Ideally, we would want to show that higher customer-to-customer trading causes bid-ask spreads to be narrower. This is difficult to show because we cannot rule out that customers select into trading bonds that are more liquid without conducting an experiment that randomly shuts off all-to-all trading for some Israeli bonds. We can, however, test whether higher customer-to-customer trading activity is negatively correlated with bid-ask spreads. To demonstrate this, we aim to control for confounding factors that could drive this negative relationship, for example, the fact that spreads are lower when trade volume is high.

We first consider the precrisis period, using data from November 1, 2019, until February 27, 2020, and regress our liquidity measure, $\log \text{BAS}_{it}$, on the total daily volume of security i traded by dealers (both with each other and with customers) and the total daily volume traded between customers. We include security- and day-fixed effects to account for unobservable security-specific and time-varying factors that affect demand and supply:

$$\log \text{BAS}_{it} = \xi_i + \xi_t + \beta_1 \log \text{dealer_volume}_{it} + \beta_2 \log \text{customer_volume}_{it} + u_{it}. \quad (5)$$

In Table 4, we find that a 1% increase in dealer trading volume corresponds to an approximately 1.8% reduction in spreads, whereas a 1% increase in customer-to-customer trading results in a reduction of around 5%.

In a second specification, we add data from the crisis period until March 13, 2020, when the Bank of Israel announced policy measures, with the results remaining robust even when the sample is narrowed

to include fewer precrisis days. To distinguish the effects during regular times (November 1, 2019, until February 27, 2020) from those during the crisis (February 28, 2020, until March 13, 2020), we introduce interaction terms between the two trade volumes and a crisis period indicator. Moreover, we include two fixed effects for each security—one for the crisis period and one for the precrisis period—to control for unobserved security-specific demand or supply shifts across these periods.

Our findings indicate that the relationship between customer-to-customer trading and liquidity remains stable during the crisis period, consistent with its behavior in normal times. Given the increase in customer volume during the crisis (Figure 9), this supports our hypothesis that all-to-all trading enhanced liquidity even amid market turmoil.

7.2. Trade Netting

Now we assess the importance of netting trades that helps to mitigate counterparty and settlement risk by conducting a counterfactual exercise. We ask how much Israeli dealer banks' gross settlement obligations would have increased if customers could only trade with dealers and netting on the Israeli exchange was like it is in the United States.

This analysis complements Fleming and Keane (2021), who quantify how much less U.S. banks' gross settlement obligations would have had decreased under central netting.¹⁴ Unlike Fleming and Keane (2021), we observe a market with central clearing and all-to-all trading. Therefore, we can (approximately) disentangle the change in dealers' gross obligations when eliminating all-to-all trading relative to U.S. market rules—while still excluding central clearing of customer trades—from the overall effect observed when transitioning from a market with central clearing and all-to-all trading to one without either central clearing or all-to-all trading.

To illustrate, consider an example with three dealers and four customers trading a security. The example mirrors the example of Fleming and Keane (2021) in Boxes A and B of the Online Appendix but with the addition of a customer-to-customer trade between customers to highlight the role of all-to-all trading. Figure 11(a) visualizes the trade flows, showing who trades with whom. For instance, dealer 1 (D1) sells 5 units to dealer 3 (D3) and buys 15 units from customer 3 (C3). Assuming all trades settle on the same day, we calculate gross settlement obligations by summing trade flows for each participant in a market with all-to-all trading but no CCP. Collectively, the dealers face gross obligations of $(10 + 5 + 10 + 15) \times 2 = 80$ among themselves and $(15 + 10 + 5 + 15) = 45$ with customers. Now assume that a CCP clears trades in the interdealer market. This allows dealers to fully net

Table 4. Customer-to-Customer Trading and the Bid-Ask-Spread

	Precrisis		Crisis vs. regular	
<i>log dealer_volume</i>	−0.018**	(0.008)	−0.018**	(0.008)
<i>log customer_volume</i>	−0.050***	(0.013)	−0.050***	(0.013)
<i>log dealer_volume</i> × <i>crisis</i>			−0.048	(0.050)
<i>log customer_volume</i> × <i>crisis</i>			+0.004	(0.048)
Observations	1,289		1,417	
Adjusted <i>R</i> ²	0.669		0.696	
Security and day fixed effects	+		+	
Security period, and day fixed effects	−		+	

Notes. Estimation findings of Regression (5) in column Precrisis. In column Crisis vs. regular, we include interaction terms with the crisis period and two security fixed effects for each security: one for the crisis period and one for regular times. In both cases, we use data for Israeli government bonds from November 2019 until March 13, 2020. The dependent variable is the daily log spread; *log dealer_volume* is the log of the total daily amount dealers trade with each other or with a customer; and *log customer_volume* is the log of the daily total amount that customers trade with each other, both measured in NIS. To avoid excluding cases when trade volume is zero, we replace $\log(0) = 0$, but the findings are robust to excluding the zeros. Indicator variable, *crisis*, is zero before February 28, 2020, and one afterward. Standard errors are clustered by security.

****p* < 0.01; ***p* < 0.05; **p* < 0.1.

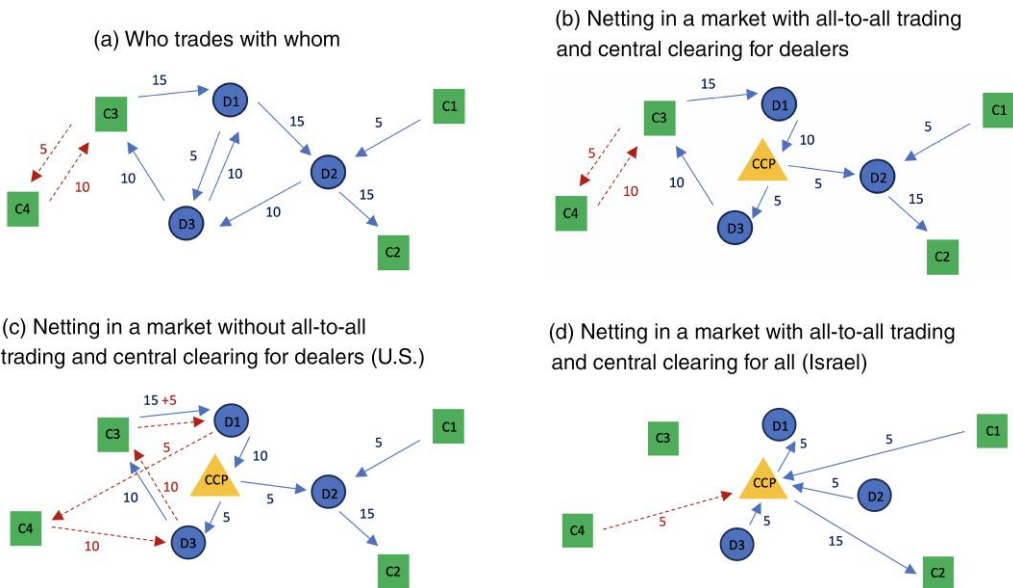
their trades with one another, as the CCP acts as the counterparty, illustrated in Figure 11(b). Because of this netting, the dealers’ gross settlement obligations decrease to 20 with the CCP while remaining at 45 with customers.

Next, in Figure 11(c), we eliminate customer-to-customer trading to align with the rules of modern OTC markets like those in the U.S. customers 3 and 4, who previously traded directly with each other, now must trade via dealers. As a result, dealers’ gross settlement obligations with customers increase to $(45 + 5 + 5 + 10 + 10) = 75$, whereas obligations with the CCP

remain at 20. In contrast, under Israeli market rules, where customers can trade directly with one another and all trades are centrally cleared (Figure 11(d)), dealers’ gross settlement obligations reduce significantly to 15.

With our data, we can approximate the changes in dealers’ gross settlement obligations when going from the status quo in Israel to U.S. rules. Because we lack information on specific trading counterparts, we cannot precisely calculate settlement obligations. Instead, as shown in Online Appendix B, we establish upper and lower bounds for these obligations.

Figure 11. (Color online) Example of Netting Benefits



Notes. Trading example between four customers (C1, C2, C3, C4) and three dealers (D1, D2, D3), and a central counterparty (CCP). Arrows indicate trade flows; for example, (a) customer 4 sells 10 units to customer 3 and buys 5 units from customer 3.

Figure 12 displays a substantial total effect on daily settlement obligations for Israeli banks when shutting off netting and all-to-all trading.

On average, daily settlement obligations in Israel were between 0.59 NIS billion and 1.06 NIS billion around the market disruptions (from January until April 2020). Under the counterfactual netting rules, these obligations rise by 1.98 NIS billion at the lower bound and by 2.46 NIS billion at the upper bound. This implies that Israeli netting rules reduced settlement obligations by 65%–81% compared with U.S. netting rules.¹⁵ Fleming and Keane (2021) find an effect of similar, yet slightly smaller magnitude—60%—for the United States.

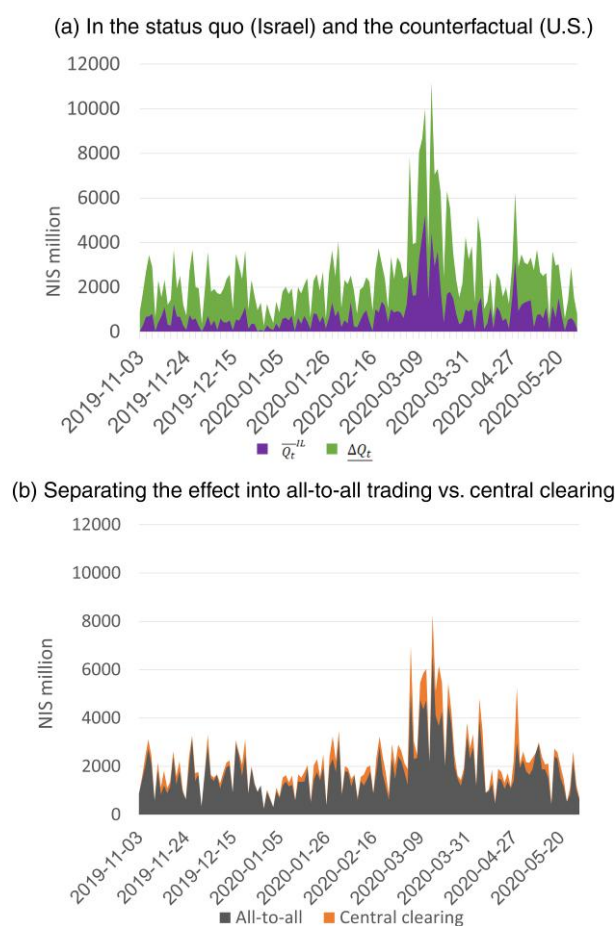
Figure 12(b) illustrates the significant impact of eliminating all-to-all trading while keeping central

clearing like it is in the United States and many other OTC markets (Figure 11(c)). It splits the dealers' daily gross obligations in our counterfactual world (at the upper bound) into obligations coming from forcing customers who would want to trade with each other to trade with dealers—customers 3 and 4 in the example—and the remaining obligations. We see that the increase in dealers' settlement obligations—caused by customers being forced to trade through dealers—can exceed the obligations arising from the inability to net trades with customers who would otherwise trade with a dealer under the current setup.

Because we do not observe the counterfactual scenario of Israel without all-to-all trading and netting, it is difficult to estimate how the (counterfactual) increase in settlement obligations would have impacted liquidity—measured, for example, by bid-ask spreads—in the Israeli market during March 2020 if there had been no exchange. However, a recent study by Duffie et al. (2023) provides valuable insights by showing that liquidity in the U.S. Treasury market is highly sensitive to dealers' balance sheet capacity. According to their findings, liquidity declines sharply once dealers reach about 50% of their capacity, as the market becomes illiquid when dealers can no longer accommodate additional trades. Beyond this threshold, illiquidity worsens rapidly, causing significant market disruptions.

Although we do not have the same data to calculate the capacity levels of Israeli dealer banks, the substantial effects shown in Figure 12—particularly the increase in customer-to-dealer trading resulting from the elimination of all-to-all trading—suggest that, without an exchange, Israeli banks might have exceeded the 50% capacity threshold. This implies that the Israeli Treasury market could have experienced severe illiquidity during March 2020, driven by dealers exhausting their balance sheet space in absence of all-to-all trading and netting. To support this conjecture, we analyze market liquidity during the financial crisis of 2007–2008, a period when dealers faced lower balance-sheet constraints, given that Basel III had not yet been implemented, and overall selling pressure was lower (Vissing-Jorgensen 2021; see Online Appendix C). In line with our conjecture, we find no significant difference in Treasury liquidity between the United States and Israel during this period.

Figure 12. (Color online) Total Dealers' Gross Obligations



Notes. (a) Upper bound of settlement obligations in Israeli's status quo ($\bar{Q}_t^{IL}$), and the increase in obligations when moving to U.S. settlement rules at the lower bound (ΔQ_t) from November 1, 2019, until June 30, 2020. (b) Decomposition of the total effect on settlement obligations at the upper bound into the effect coming from all-to-all trading versus the additional effect coming from netting. All obligations are expressed in NIS million and stacked on top of each other in both graphs. See Online Appendix B for details on how we compute all bounds.

7.3. Take Away

The evidence presented suggests that an exchange for government bonds in the United States and other countries could promote liquidity in a crisis because it allows investors to provide liquidity for one another and because it facilitates the netting of trades. Importantly, these two factors are complementary in that a

large part of the netting effect comes from the fact that fewer trades go via dealers when investors can trade with one another.

Taken together, our findings highlight the fact that trading mechanisms and financial regulations or policies that affect dealer balance sheets are intertwined. Thus, they must be evaluated together to avoid future market distress, especially in an era in which dealer banks' balance sheets are crowded by unprecedented large amounts of government debt.

8. Conclusion

This paper analyzes whether and why shifting the trading of government bonds onto an exchange can protect market liquidity against negative shocks in times of crisis. We leverage the institutional feature whereby only the Israeli bond market trades on exchange. We first provide descriptive evidence that suggests that having an exchange may foster liquidity and then test this hypothesis via difference-in-differences analyses. We show that Israeli dealer banks and investors leaned against the wind in March 2020 and conduct a counterfactual to highlight the importance of trade netting and all-to-all trading.

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Endnotes

¹ Proposals that move toward full central clearing have already been voted for. For instance, on September 14, 2022, the Securities and Exchange Commission proposed that all trades made on automated and anonymous interdealer broker platforms should go through clearinghouses. Earlier it had proposed to force market participants trading more than \$25 billion a month in Treasuries to register as dealers (see <https://www.ft.com/content/cffb3c3c-2c06-42c3-92e3-839a1e9ff5e8>, accessed on February 20, 2023).

² Extensive theoretical literature studies the fragmentation or centralization of financial markets (Glosten 1994; Glode and Opp 2019; Wittwer 2020, 2021; Baldauf and Mollner 2021; Chen and Duffie 2021; Rostek and Yoon 2021; Budish et al. 2024).

³ The Israeli market may differ from those in other countries due to its unique history. Abudy and Wohl (2018) explain that the TASE was established in 1935 when market conditions were different, with low demand and supply leading to a single daily auction. As the market grew, participants became accustomed to trading on the exchange, preventing the development of an OTC market. In

contrast, other countries where markets were more liquid, for example, the United States, saw bond trading shift from exchanges to OTC markets as institutional investors sought better deals, leading to a decline in exchange liquidity by 1940 (Biais and Green 2019). For more on Israel, see Online Appendix A.

⁴ The Korean Ministry of Economy and Finance acknowledges that “[s]imilar to other bond markets, the OTC market represents the dominant volume of bond and note trading” (South Korean Ministry of Economy and Finance 2024). This stands in contrast to Israel, where the vast majority of government bonds trade on the exchange.

⁵ Tier 1 capital consists primarily of common stock and disclosed reserves (or retained earnings) but may also include nonredeemable noncumulative preferred stock; leverage exposure includes the total notional of all cash and repo transactions of all securities, including government bonds.

⁶ For an analysis of the effectiveness of these interventions, see Nathan (2020) and chapter 3 of the Bank of Israel's 2020 yearly report.

⁷ We use daily rather than more high-frequency data to reduce differences across countries that arise because of different time zones.

⁸ Because the estimates are large, the standard log approximation is not accurate.

⁹ For more discussion of how government bonds may differ from other types of securities, see, for example, Nagel (2016), who shows that government bonds are near money assets, or Boudoukh et al. (2021), who show that in previous crisis periods, investors sold more liquid government bonds.

¹⁰ Online Appendix Table A6 reports the estimation coefficients when estimating the fixed effect regression (3) for each country separately rather than pooling across countries and absorbing country differences by including country-day fixed effects. Because many time-varying factors affect demand and supply of government bonds and stocks, we expect the size of the effect to vary across countries. We find that it ranges approximately between 10% and 30% for all countries, excluding the United States and Poland. In the United States, the effect is the largest with 49%.

¹¹ This implies omitting the postvariable, and for the United States, where the same security does not trade on and off the exchange, also the exchange-variable given perfect collinearity with the fixed effects.

¹² Sophisticated investors tried to substitute into the futures market and sell futures rather than Treasuries until prices in the cash market were more favorable. They were successful only to the extent that they found a counterparty in the repo market. In addition, there was an opposing effect. Hedge funds that had taken a short position in Treasury futures and a long position in Treasury securities before the crisis unwound their position when futures became more expensive relative to cash treasuries (Barth and Kahn 2020, Schrimpf et al. 2020). Thus, they sold treasuries and bought futures.

¹³ In regular times, the spread on the exchange is larger than off the exchange. This is probably because investors that trade large amounts OTC are more similar to dealers and therefore trade at a lower spread than the average investor on the exchange.

¹⁴ Like Fleming and Keane (2021), we focus on the uncleared portion of intermediating cash trades and abstract from the outright long and repo positions, which might be balance-sheet intensive. This implies that we abstract from other reasons institutions may pull back from intermediating cash trades, such as margins in a CCP, unsecured credit to intermediate unsettled trades, and risk management practices.

¹⁵ The lower bound of the percentage reduction is $1 - 1/(1.98/1.06 + 1) = 0.65$, and the upper bound is $1 - 1/(2.46/0.59 + 1) = 0.81$.

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